

JANASANKHYA

real data → CTGAN → cited priors → AI audit

A synthetic population of India's informal workers — built to test whether AI serves them.

45,424

real workers
IHDS-II microdata

10,000

synthetic workers
from a real CTGAN

0.1463

mean KS
fidelity (lower=better)

0.0

memorised rows
no real person copied

22 pts

language gap
in AI welfare advice

How it is built

LAYER 1 — CTGAN

learns the real joint distribution of 45,424 IHDS-II informal workers → 10,000 synthetic workers

LAYER 2 — STRUCTURAL PRIORS

imputes wage-violation, rights-awareness, AI-usability from cited reports (Jan Sahas, ILO, GSMA)

LAYER 3 — LLM PERSONAS

each worker reasons in first person; cross-validates Layer 2 (agreement within ±1 for 100% of workers)

LAYER 4 — AI AUDIT

asks each worker's welfare question to a live AI in 5 languages; scores vs real scheme-eligibility

THE FINDING — AI welfare advice degrades by language same worker, same question (n=50, 95% CI)

